E016

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# Assignment 7

1. $X(t) = \cos \omega t$

$\omega \sim U[0,1]$

$$E[X(t)] = \int_{-\infty}^{\infty} X(t) \cdot f(x)\, dx$$

$$f(\omega) = 1$$

$$E[X(t)] = \int_0^1 \cos \omega t \, d\omega = \left[\frac{\sin \omega t}{t}\right]_0^1$$

$$= \frac{\sin \cancel{\omega} t}{t} - \frac{\sin(0)}{t} = \frac{\sin(t)}{t}$$

∴ Not a constant, but a function of t

∴ Not stationary

2. $X(t) = A \cdot \cos(\omega_0 t + \theta)$

$\theta \sim U[0, \pi]$

$$E[X(t)] = \frac{1}{\pi} \int_0^{\pi} A \cdot \cos(\omega_0 t + \theta)\, d\theta$$

$$= \frac{A}{\pi}\left[\sin(\omega_0 t + \theta)\right]_0^{\pi} \rightarrow \frac{A}{\pi}[\sin(\omega_0 t + \pi) - \sin(\omega_0 t)]$$

$$= \frac{A}{\pi}[-\sin(\omega_0 t) - \sin(\omega_0 t)]$$

$$= \frac{-2A}{\pi}(\sin(\omega_0 t))$$

∴ Not a constant but a function of t

∴ $x(t)$ is not stationary

3. $E[X(t)] = E[A\cos(\omega_0 t + \theta)]$

$\theta \sim U[0, \pi/2]$

$$E[X(t)] = \frac{2}{\pi}\int_0^{\frac{\pi}{2}} A\cos(\omega_0 t + \theta)\, d\theta$$

$$= \frac{2A}{\pi}\Big[\sin(\omega_0 t + \theta)\Big]_0^{\pi/2}$$

$$= \frac{2A}{\pi}\Big[\sin(\omega_0 t + \pi/2) - \sin(\omega_0 t)\Big]$$

$$= \frac{2A}{\pi}\big(\cos(\omega_0 t) - \sin(\omega_0 t)\big)$$

$\therefore$ Not a constant, but function of t

$\therefore$ $X(t)$ isn't stationary.

4. $X(t) = A\cos(\omega_0 t)$

$A \sim U[0,1]$

$$E[X(t)] = \int_0^1 A\cdot\cos(\omega_0 t)\cdot dA$$

$$= \cos(\omega_0 t)\left[\frac{A^2}{2}\right]_0^1 \rightarrow \frac{1}{2}\cos(\omega_0 t)$$

$\therefore$ Not a constant, but a function of t

$\therefore$ $X(t)$ isn't stationary.

5. $X(t) = 10\cos(100t + \theta)$

$\theta \sim U[-\pi, \pi]$

$$E[X(t)] = \frac{1}{2\pi}\int_{-\pi}^{\pi} 10\cdot\cos(100t + \theta)\, d\theta$$

$$= \frac{5}{\pi}\Big[\sin(100t + \theta)\Big]_{-\pi}^{\pi}$$

$$= \frac{5}{\pi}\left[\sin(\pi+100t) + \sin(\pi-100t)\right]$$

$$= \frac{5}{\pi}\left[-\sin(\cancel{100t}) + \sin(\cancel{100t})\right]$$

$$= 0 \rightarrow \text{constant.}$$

$$R_x(\tau) = E[x(t)\cdot x(t+\tau)]$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} 10\cos(100t+\theta)\cdot 10\cos(100t+\theta+100\tau)\,d\theta$$

$$= \cancel{\frac{50}{\pi}\left[\sin(100t+\theta)\right.}$$

$$= \frac{50}{\pi}\int_{-\pi}^{\pi} \cos(100t+\theta)\cdot\cos(100t+\theta+100\tau)\,d\theta$$

$$= \frac{25}{\pi}\int_{-\pi}^{\pi} \cos(200t+2\theta+\tau) + \cos(-100\tau)\,d\theta$$

$$= \frac{25}{\pi}\left[\frac{\sin(200t+2\theta+\tau)}{2} + \cos(100\tau)\cdot\theta\right]_{-\pi}^{\pi}$$

$$= \frac{25}{\pi}\left[\frac{\sin(200t+100\tau+2\pi)}{2} + \pi\cos(100\tau) + \pi\cos(100\tau) + \frac{\sin(2\pi-(200t+100\tau))}{2}\right]$$

$$= \frac{25}{\cancel{\pi}}\left[2\cancel{\pi}\cos(100\tau) + \frac{\sin(\cancel{200t}+100\tau)}{2} - \frac{\sin(\cancel{200t}+100\tau)}{2}\right]$$

$$= 50\cos(100\tau)$$

$\therefore$ Function of $\tau$

$\therefore$ It is WSS

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6. $x(t) = \cancel{A}\sin(\omega_0 t + Y)$

$Y \sim U[0, 2\pi]$

$$f(Y) = \frac{1}{2\pi}$$

- $$E[x(t)] = \frac{1}{2\pi}\int_0^{2\pi} \cancel{A}\cdot\sin(\omega_0 t + Y)\, dY$$

$$= \frac{\cancel{A}1}{2\pi}\Big[-\cos(\omega_0 t + Y)\Big]_0^{2\pi} = \frac{\cancel{A}1}{2\pi}\Big[-\cos(2\pi + \omega_0 t) + \cos(\omega_0 t)\Big]$$

$$= \frac{\cancel{A}1}{2\pi}\Big[-\cancel{\cos}(\omega_0 t) + \cancel{\cos}(\omega_0 t)\Big] = 0$$

- $$R_x(\tau) = E[x(t)\cdot x(t+\tau)]$$

$$= E\Big[\cancel{A}\sin(\omega_0 t + Y)\cdot \cancel{A}\,\overset{\sin}{\cancel{\cos}}(\omega_0(t+\tau) + Y)\Big]$$

$$= \frac{1\cancel{A^2}}{2\pi}\int_0^{2\pi} \sin(\omega_0 t + Y)\cdot \overset{\sin}{\cancel{\cos}}(\omega_0 t + \omega_0\tau + Y)\, dY$$

$$= \frac{1\cancel{A^2}}{4\pi}\int_0^{2\pi} \cos(\cancel{\omega_0 t} + \cancel{Y} - \cancel{\omega_0 t} - \omega_0\tau - \cancel{Y}) - \cos(\omega_0 t + Y + \omega_0 t + \omega_0\tau + Y)\, dY$$

$$= \frac{1\cancel{A^2}}{4\pi}\int_0^{2\pi} \cos(-\omega_0\tau) - \cos(2\omega_0 t + 2Y + \omega_0\tau)\, dY$$

$$= \frac{1\cancel{A^2}}{4\pi}\Big[\cancel{\sin}\cos(\omega_0 t)\cdot Y - \cancel{\frac{\sin(2\omega_0 t + 2Y + \omega_0\tau)}{2}}\Big]_0^{2\pi}$$

$$= \frac{1\cancel{A^2}}{4\pi}\Big[\cos(\omega_0 \overset{\tau}{\cancel{t}})\cdot 2\pi - \cancel{\frac{\sin((2\omega_0 t + \omega_0\tau) + 4\pi)}{2}} + \cancel{\frac{\sin(2\omega_0 t + \omega_0\tau)}{2}}\Big]$$

$$R_x(\tau) = \frac{1\cancel{A^2}}{2}\cos(\omega_0 \overset{\tau}{\cancel{t}})$$ → As mean is constant and $R_x$ depends on $\tau$ → WSS

7. $X(t) = A\sin(\omega_0 t + \theta)$

$\theta \sim U[-\pi, \pi]$

$$E[X(t)] = \frac{1}{2\pi}\int_{-\pi}^{\pi} A\cdot\sin(\omega_0 t + \theta)\, d\theta$$

$$= \frac{A}{2\pi}\int_{-\pi}^{\pi} \sin(\omega_0 t + \theta)\, d\theta$$

$$= \frac{-A}{2\pi}\Big[\cos(\omega_0 t + \theta)\Big]_{-\pi}^{\pi}$$

$$= \frac{-A}{2\pi}\Big[\cos(\omega_0 t + \pi) - \cos(\omega_0 t - \pi)\Big]$$

$$= \frac{-A}{2\pi}\Big[\cos(\omega_0 t + \pi) + \cos(\pi - \omega_0 t)\Big]$$

$$= \frac{-A}{2\pi}\Big[\cos(\omega_0 t) - \cos(\omega_0 t)\Big] = 0 \rightarrow \text{constant}$$

$$R_X(\tau) = E[X(t)\cdot X(t+\tau)]$$

$$= E[A\sin(\omega_0 t + \theta)\cdot A\sin(\omega_0(t+\tau) + \theta)]$$

$$= \frac{A^2}{2\pi}\int_{-\pi}^{\pi} \sin(\omega_0 t + \theta)\cdot \sin(\omega_0 t + \omega_0\tau + \theta)\, d\theta$$

$$= \frac{A^2}{4\pi}\int_{-\pi}^{\pi} \cos(-\omega_0\tau) - \cos(2\omega_0 t + 2\theta + \omega_0\tau)\, d\theta$$

$$= \frac{A^2}{4\pi}\left[\cancel{\sin\theta}\ \theta\cos(\omega_0\tau) - \frac{\sin(2\omega_0 t + \omega_0\tau + 2\theta)}{2}\right]_{-\pi}^{\pi}$$

$$= \frac{A^2}{4\pi}\Big[\pi\cos(\omega_0 t) - \frac{\sin(2\pi + (2\omega_0 t + \omega_0\tau))}{2} +$$

$$\pi\cos(\omega_0 t) + \frac{\sin(-2\pi + 2\omega_0 t + \omega_0\tau)}{2}\Big]$$

$$= \frac{A^2}{2\cancel{4}\cancel{\pi}}\Big[2\cancel{\pi}\cos(\omega_0 t) - \frac{\sin(2\omega_0 t + \omega_0\tau)}{\cancel{2}} + \frac{\sin(2\omega_0 t + \omega_0\tau)}{\cancel{2}}$$

$= \frac{A^2}{2} \cos(\omega_0 \tau) \rightarrow$ function of $\tau$

$\therefore$ WSS

8. $X(t) = \sin(\omega_0 t + \theta)$

$\theta \sim U[0, 2\pi]$

$f(\theta) = \frac{1}{2\pi}$

$$E[X(t)] = \frac{1}{2\pi} \int_0^{2\pi} \sin(\omega_0 t + \theta)\, d\theta$$

$$= \frac{1}{2\pi} \left[ -\cos(\omega_0 t + \theta) \right]_0^{2\pi} = \frac{1}{2\pi} \left[ -\cos(2\pi + \omega_0 t) + \cos(\omega_0 t) \right]$$

$$= \frac{1}{2\pi} \left[ -\cancel{\cos(\omega_0 t)} + \cancel{\cos(\omega_0 t)} \right] = 0$$

$$R_X(\tau) = E[X(t) \cdot X(t+\tau)]$$

$$= E[\sin(\omega_0 t + \theta) \cdot \sin(\omega_0 (t+\tau) + \theta)]$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \sin(\omega_0 t + \theta) \cdot \sin(\omega_0 t + \omega_0 \tau + \theta)\, d\theta$$

$$= \frac{1}{4\pi} \int_0^{2\pi} \cos(-\omega_0 \tau) - \cos(2\omega_0 t + 2\theta + \omega_0 \tau)\, d\theta$$

$$= \frac{1}{4\pi} \left[ \cos(\omega_0 \tau) \cdot 2\pi - \cancel{\frac{\sin(4\pi + (2\omega_0 t + \omega_0 \tau))}{2}} + \cancel{\frac{\sin(2\omega_0 t + \omega_0 \tau)}{2}} \right]$$

$$R_X(\tau) = \frac{1}{2} \cos(\omega_0 \tau)$$

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$\therefore$ As mean is constant & $R_X$ depends on $\tau$

$\therefore$ WSS.

9. $X(t) = A e^{(\omega_0 t + \theta)}$

$E[A] = 0$

$E[X(t)] = E[A e^{(\omega_0 t + \theta)}]$
$= E[A] \cdot E[e^{(\omega_0 t + \theta)}]$
$= 0$

$R_x(\tau) = E[A e^{(\omega_0 t + \theta)} \cdot A e^{(\omega_0 t + \omega_0 \tau + \theta)}]$
$= E[A^2] \cdot E[e^{2\omega_0 t + 2\theta + \omega_0 \tau}]$
$= E[A^2] \cdot E[e^{2\omega_0 t}] \cdot E[e^{2\theta}] \cdot E[e^{\omega_0 \tau}]$

$\therefore$ Not WSS

10. $R_{x,x}(\tau) = A e^{-\alpha|\tau|}$
$Y = X(7) - X(4)$
$E[Y^2] = ?$
$E[Y^2] = E[(X(7) - X(4))^2]$
$= E[X^2(7) + X^2(4) - 2 \cdot X(7) \cdot X(4)]$
$= E[X^2(7)] + E[X^2(4)] - 2E[X(7) \cdot X(4)]$
$= \cancel{E[X^2(7)] + E[X^2(4)] - 2}$

As $\tau = t_1 - t_2$

$= R_{x,x}(0) + R_{x,x}(0) - 2 \cdot R_{x,x}(3)$
$= A + A - 2(Ae^{-3\alpha})$
$= 2A - 2Ae^{-3\alpha}$

$\boxed{E[Y^2] = 2A(1 - e^{-3\alpha})}$

11. $X(t) = X_1(t) + t \cdot X_2(t)$

$E(X_1) = a$, $Var(X_1) = \sigma_1^2$

$E(X_2) = b$, $Var(X_2) = \sigma_2^2$

$E[X(t)] = E[X_1(t) + t \cdot X_2(t)]$

$= E[X_1(t)] + t \cdot E[X_2(t)]$

$\boxed{E[X(t)] = a + t \cdot b}$

$R_X(t_1, t_2) = E[X(t_1) \cdot X(t_2)]$

$= E[X_1(t_1) + t_1 X_2(t_1) \cdot X_1(t_2) + t_2 X_2(t_2)]$

But as $X_1, X_2$ are independent RV

$X_1(t_1) = X_1(t_2) = X_1$

$X_2(t_1) = X_2(t_2) = X_2$

$R_X(t_1, t_2) = E[X_1 + t_1 \cdot X_2 \cdot X_1 + t_2 X_2]$

$= E[X_1^2 + t_2 X_1 X_2 + t_1 X_1 X_2 + t_1 t_2 X_2^2]$

$= E[X_1^2] + (t_1 + t_2) E[X_1 \cdot X_2] + t_1 t_2 E[X_2^2]$

$\boxed{R_X(t_1, t_2) = (\sigma_1^2 + a^2) + (t_1 + t_2) \cdot ab + t_1 t_2 (\sigma_2^2 + b^2)}$

$C_X(t_1, t_2) = R_X(t_1, t_2) - E[X(t_1)] \cdot E[X(t_2)]$

~~$= R_X(t_1, t_2) - E[X$~~

$E[X(t_1)] = a + t_1 b$

$E[X(t_2)] = a + t_2 b$

$E[X(t_1)] \cdot E[X(t_2)] = a^2 + t_2 \cdot ab + t_1 \cdot ab + b^2 t_1 t_2$

$= a^2 + ab(t_1 + t_2) + b^2 (t_1 t_2)$

$C_X(t_1, t_2) = \sigma_1^2 + \cancel{a^2} + ab(\cancel{t_1 + t_2}) + t_1 t_2 (\sigma_2^2 + \cancel{b^2})$

$- \cancel{a^2} - \cancel{ab(t_2)} - \cancel{ab(t_1)} - \cancel{t_1 t_2 (b^2)}$

$\boxed{C_X(t_1, t_2) = \sigma_1^2 + t_1 t_2 (\sigma_2^2)}$

- Our mean is $a+bt$ → depends on 't', unless 'b' is 0.
- $R_x$ depends on $t_1 + t_2$ and $t_1 t_2$

∴ Process isn't stationary.

12. $x(t) = \sin(\omega t + Y)$

$Y \sim U[-\pi, \pi]$

$$f(Y) = \frac{1}{2\pi}$$

$$E[X(t)] = E[\sin(\omega t + Y)]$$

$$= \int_{-\pi}^{\pi} \frac{1}{2\pi} \cdot \sin(\omega t + Y)\, dY$$

$$= \frac{1}{2\pi} \Big[-\cos(\omega t + Y)\Big]_{-\pi}^{\pi}$$

$$= \frac{1}{2\pi} \Big[-\cos(\omega t + \pi) + \cos(\omega t - \pi)\Big]$$

$$= \frac{1}{2\pi} \Big[\cancel{\cos}\ \cancel{\cos(\omega t)} - \cancel{\cos(\pi - \omega t)}\Big]$$

$$= \frac{1}{2\pi} \Big[\cancel{\cos(\omega t)} + \cancel{\cos(\omega t)}\Big]$$

$$= \frac{\cos(\omega t)}{\pi}$$

$$\cos(-(\pi - \omega t)) = \cos(\pi - \omega t)$$

$$= \frac{1}{2\pi} \Big[\cos(\omega t) + \cos(\pi - \omega t)\Big]$$

$$= \frac{1}{2\pi} \Big(\cos(\cancel{\omega t}) - \cancel{\cos}(\omega t)\Big)$$

$$= 0$$

$$R_x(\tau) = E[X(t) \cdot X(t+\tau)]$$

$$= E[\sin(\omega t + Y) \cdot \sin(\omega t + \omega\tau + Y)]$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} \sin(\omega t + Y) \cdot \sin(\omega t + \omega\tau + Y) \cdot dY$$

$$= \frac{1}{4\pi}\int_{-\pi}^{\pi} \cos(\omega\tau) - \cos(2\omega t + \cancel{2}\omega\tau + 2Y)\, dY$$

$$= \cancel{\frac{1}{4\pi}}\ \cancel{\sin(\omega}$$

$$= \frac{1}{4\pi}\left[Y\cos(\omega\tau) - \frac{\sin(2\omega t + \omega\tau + 2Y)}{\cancel{4}\,2}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{4\pi}\left[\pi\cos(\omega\tau) - \frac{\sin(2\pi + (2\omega t + \omega\tau))}{2} + \pi\cos(\omega\tau) - \frac{\sin(2\pi - (2\omega t + \omega\tau))}{2}\right]$$

$$= \frac{1}{4\pi}\left[2\pi\cos(\omega\tau) - \cancel{\frac{\sin(2\omega t + \omega t)}{2}} + \cancel{\frac{\sin(2\omega t + \omega\tau)}{2}}\right]$$

$$= \frac{1}{2}\cos(\omega\tau)$$

$\therefore$ As mean $= 0 \rightarrow$ constant

$\therefore$ $R_x$ is a func of $\tau$

$\therefore$ ~~Stationary~~ WSS //

13. $X(t) = A\cos t + B\sin t$

$E[A] = \sum p_i x_i$

$= \frac{1}{2}(-1) + \frac{1}{2}(1) = 0$

$E[B] = \frac{1}{2}(-1) + \frac{1}{2}(1) = 0$

$E[A] = E[B] = 0$

$E[X(t)] = E[A\cos t + B\sin t]$

$= \cos t \cdot E[A] + \sin t \cdot E[B]$

$= 0 \rightarrow$ constant.

$R_X(\tau) = E[X(t) \cdot X(t+\tau)]$

$= E[A\cos t + B\sin t \cdot A\cos(t+\tau) + B\sin(t+\tau)]$

$= E[A^2\cos t \cdot \cos(t+\tau) + AB\cos t \cdot \sin(t+\tau) + AB\sin t \cdot \cos(t+\tau) + B^2\sin t \cdot \sin(t+\tau)]$

$= \cos t \cdot \cos(t+\tau) \cdot E[A^2] + \cos t \cdot \sin(t+\tau) \cdot E[AB] + \sin t \cdot \cos(t+\tau) \cdot E[AB] + \sin t \cdot \sin(t+\tau) \cdot E[B^2]$

$E[A^2] = \sum p_i x_i^2 = E[B^2]$

$= \frac{1}{2}(1) + \frac{1}{2}(1) = 1$

$E[AB] = E[A] \cdot E[B]$

$= 0$

$R_X(\tau) = \cos t \cdot \cos(t+\tau) + \sin t \cdot \sin(t+\tau)$

$= \cos(t - (t+\tau))$

$= \cos(-\tau)$

$= \cos(\tau) \rightarrow$ function of $\tau$

$\therefore$ WSS

14. $x(t) = A\cos(2\pi t)$

$E[x(t)] = E[A\cos(2\pi t)]$
$= \cos(2\pi t) \cdot E[A] \rightarrow 0$ if $E[A] = 0$

~~$R_x(\tau) = E[x(t) \cdot x(t+\tau)]$~~
~~$= E[A\cos(2\pi t) \cdot A\cos(2\pi t + 2\pi(\tau))]$~~

$\because$ As $E[A]$ isn't a constant
$\therefore$ Not stationary.

15. $E[X_1] = 1$, $V[X_1] = \frac{(2-0)^2}{12} = \frac{1}{3}$

$E[X_2] = 1.5$, $V[X_2] = \frac{(3-0)^2}{12} = \frac{3}{4}$

$E[x(t)] = E[1+1.5]$
$= 2.5$

$C_x(t_1, t_2) = V[X_1] + V[X_2]$
$= \frac{13}{12}$

$R_x(t_1, t_2) = C_x + (E[x])^2$
$= \frac{13}{12} + (2.5)^2$

$= 7.33$

$\therefore$ The process is WSS

RVP
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